

COURSE CODE/TITLE:	CPE 212 / Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	7/23/2026	DATE SUBMITTED:	7/23/2026
NAME:	Galan, Irick Marvin A.	INSTRUCTOR:	Engr. Jimlord Quejado

ACTIVITY NO. 3
Install SSH Server on CentOS or RHEL 8

1. PROCEDURE OUTPUT

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here:
http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/


CentOS

Table of Contents

CentOS Stream

Continuously delivered distro that tracks just ahead of Red Hat Enterprise Linux (RHEL) RHEL. For anyone interested in participating and collaborating in the RHEL ecosystem,

10

9

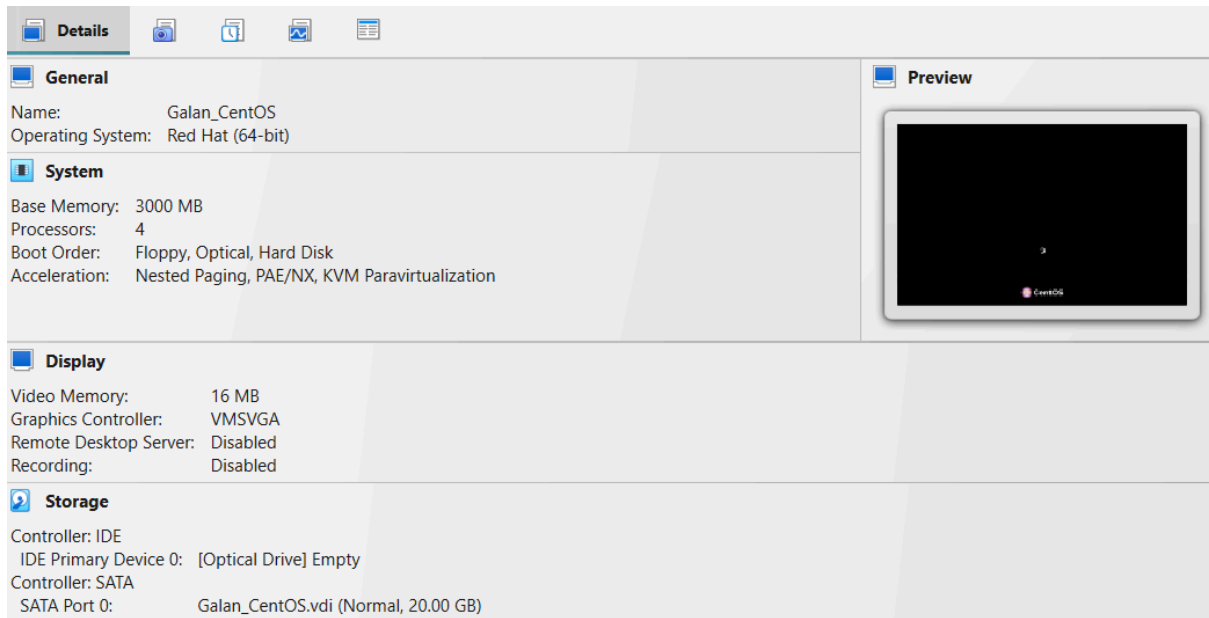
Architecture	ISOs	RPMs
x86_64	Mirrors	Mirrors
ARM64 (aarch64)	Mirrors	Mirrors
IBM Power (ppc64le)	Mirrors	Mirrors
IBM Z (s390x)	Mirrors	Mirrors

Documentation

[Release Announcement](#)
[Release Notes](#)

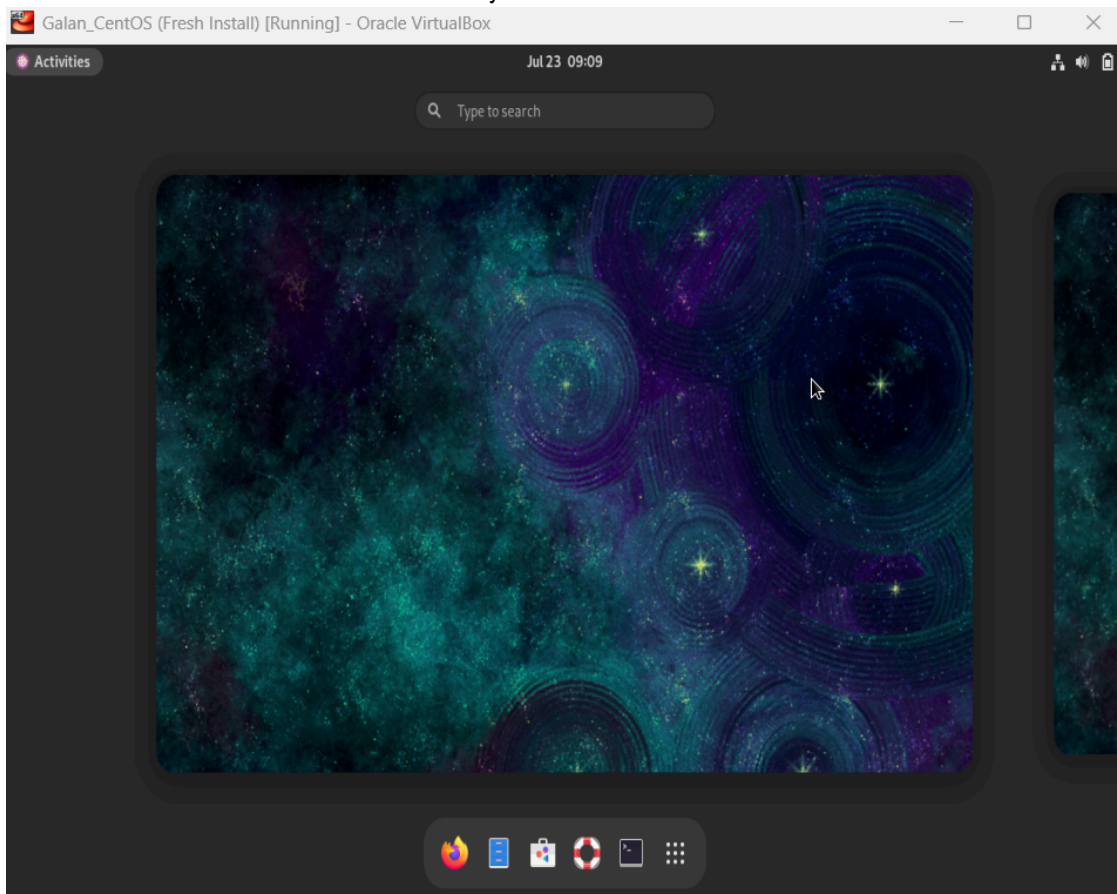
End-of-life

2. Create a VM machine with 2 Gb RAM and 20 Gb HD.



3. Install the downloaded image.

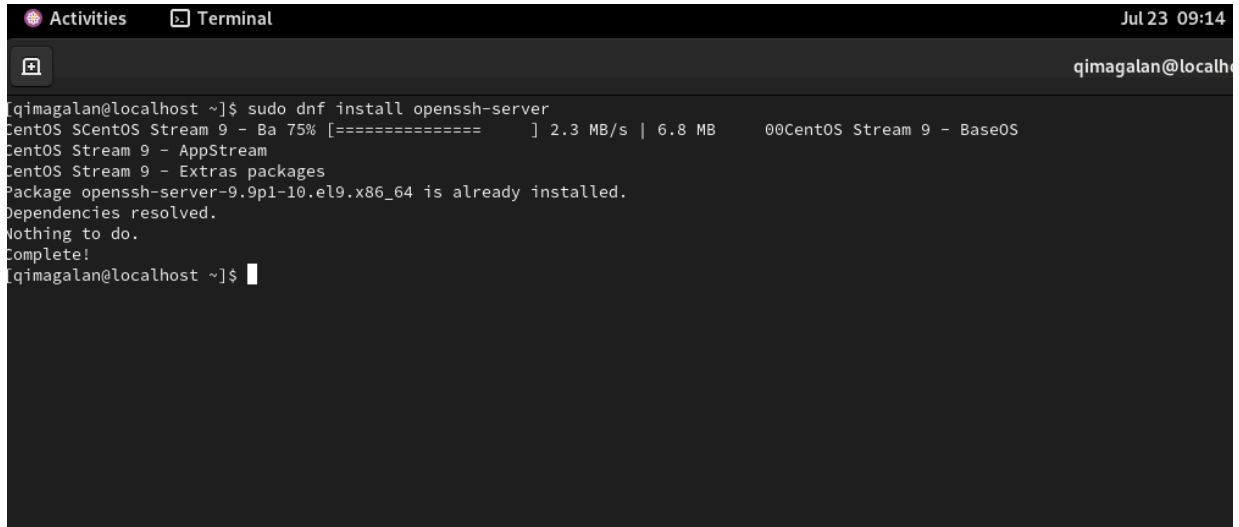
4. Show evidence that the OS was installed already.



Task 2: Install the SSH server package openssh

1. Install the ssh server package openssh by using the dnf command:

\$ dnf install openssh-server



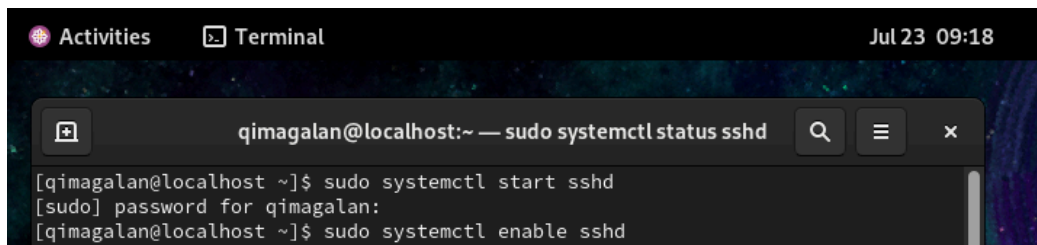
```

qimagalan@localhost ~]$ sudo dnf install openssh-server
CentOS SCentOS Stream 9 - Ba 75% [=====] 2.3 MB/s | 6.8 MB    00CentOS Stream 9 - BaseOS
CentOS Stream 9 - AppStream
CentOS Stream 9 - Extras packages
Package openssh-server-9.9p1-10.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
qimagalan@localhost ~]$
  
```

2. Start the sshd daemon and set to start after reboot:

\$ systemctl start sshd

\$ systemctl enable sshd

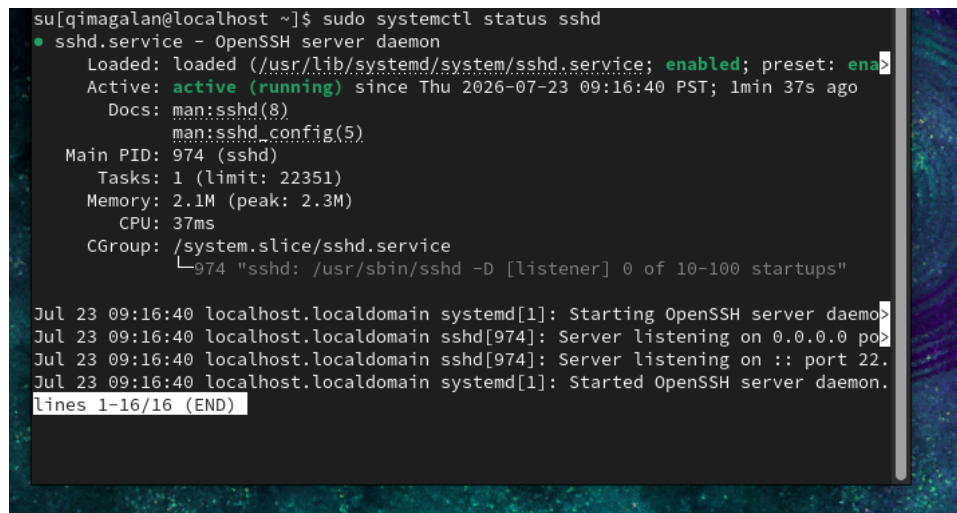


```

qimagalan@localhost:~ — sudo systemctl status sshd
[qimagalan@localhost ~]$ sudo systemctl start sshd
[sudo] password for qimagalan:
[qimagalan@localhost ~]$ sudo systemctl enable sshd
  
```

3. Confirm that the sshd daemon is up and running:

\$ systemctl status sshd



```

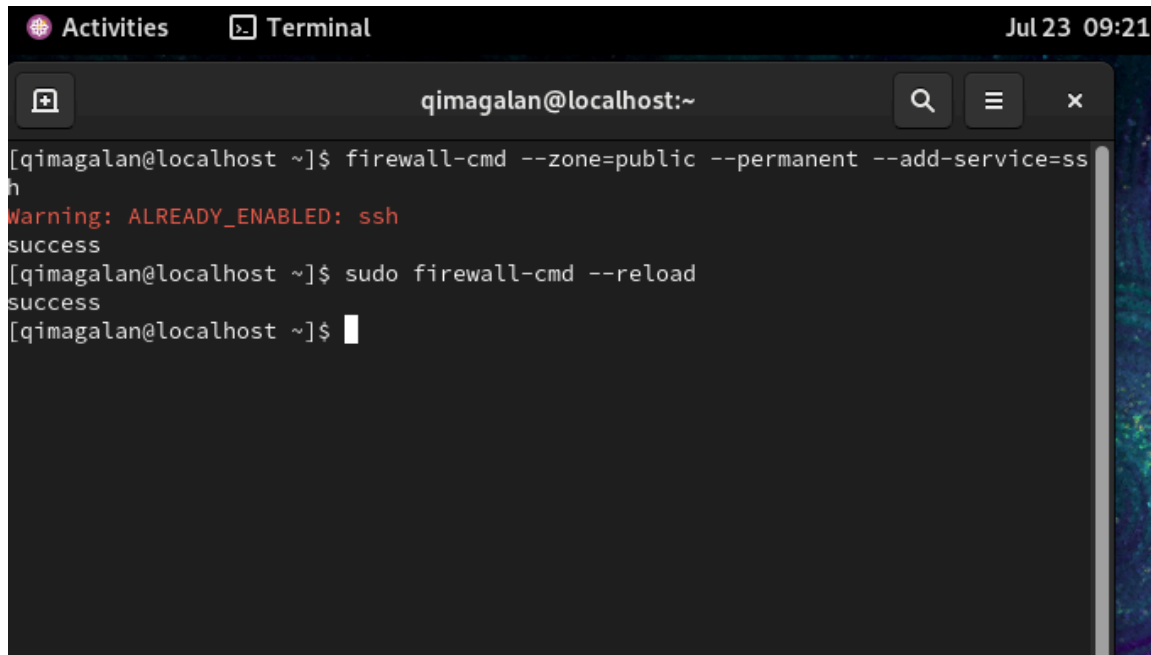
su[qimagalan@localhost ~]$ sudo systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: ena
   Active: active (running) since Thu 2026-07-23 09:16:40 PST; 1min 37s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 974 (sshd)
      Tasks: 1 (limit: 22351)
    Memory: 2.1M (peak: 2.3M)
       CPU: 37ms
    CGroup: /system.slice/ssh.service
           └─974 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 09:16:40 localhost.localdomain systemd[1]: Starting OpenSSH server daemon.
Jul 23 09:16:40 localhost.localdomain sshd[974]: Server listening on 0.0.0.0 po
Jul 23 09:16:40 localhost.localdomain sshd[974]: Server listening on :: port 22.
Jul 23 09:16:40 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
lines 1-16/16 (END)
  
```

4. Open the SSH port 22 to allow incoming traffic:

\$ firewall-cmd --zone=public --permanent --add-service=ssh

\$ firewall-cmd --reload



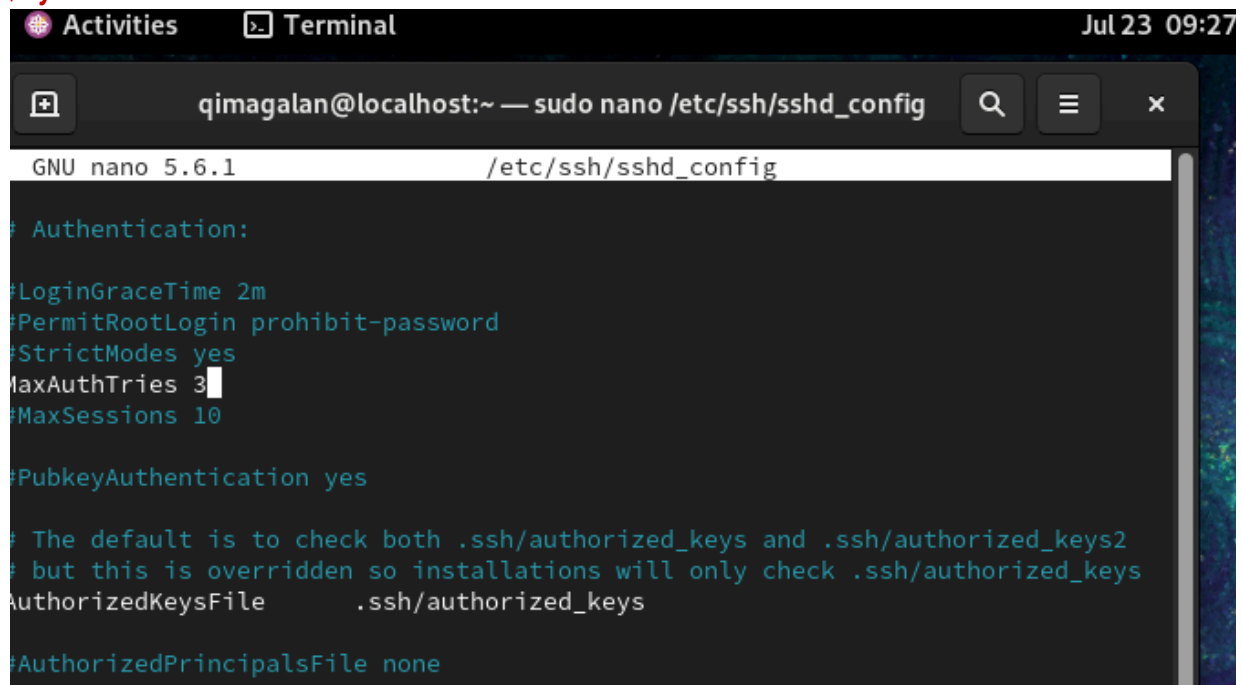
```

qimagalan@localhost:~
[qimagalan@localhost ~]$ firewall-cmd --zone=public --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[qimagalan@localhost ~]$ sudo firewall-cmd --reload
success
[qimagalan@localhost ~]$

```

5. Locate the ssh server man config file `/etc/ssh/sshd_config` and perform custom configuration. Every time you make any change to the `/etc/ssh/sshd-config` configuration file reload the sshd service to apply changes:

\$ systemctl reload sshd



```

qimagalan@localhost:~ — sudo nano /etc/ssh/sshd_config
GNU nano 5.6.1 /etc/ssh/sshd_config

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
MaxAuthTries 3
#MaxSessions 10

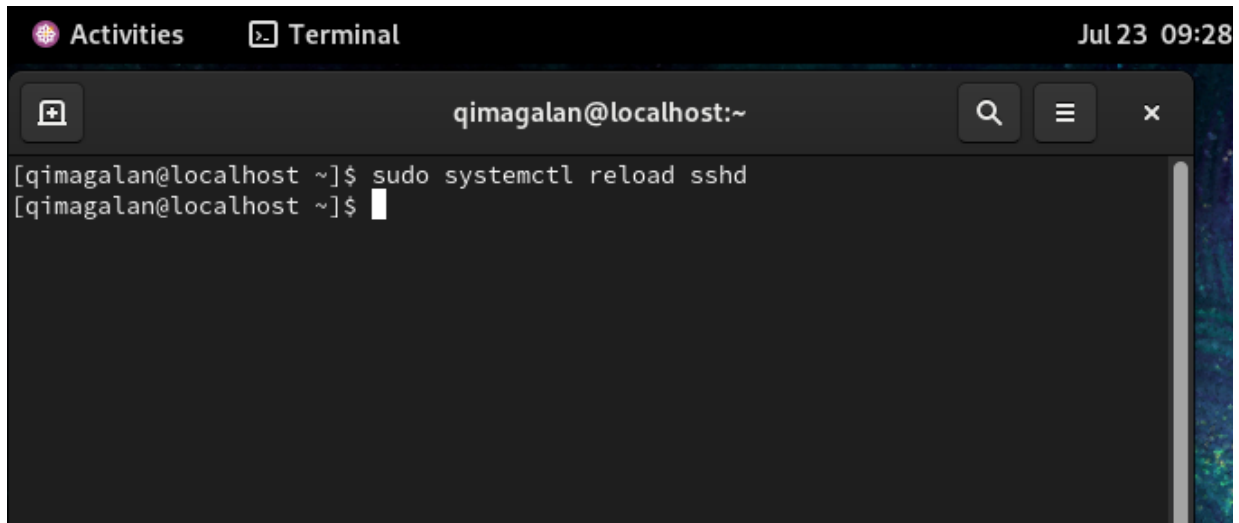
#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile .ssh/authorized_keys

#AuthorizedPrincipalsFile none

```

Changed the maximum authentication tries to 3 attempts



```

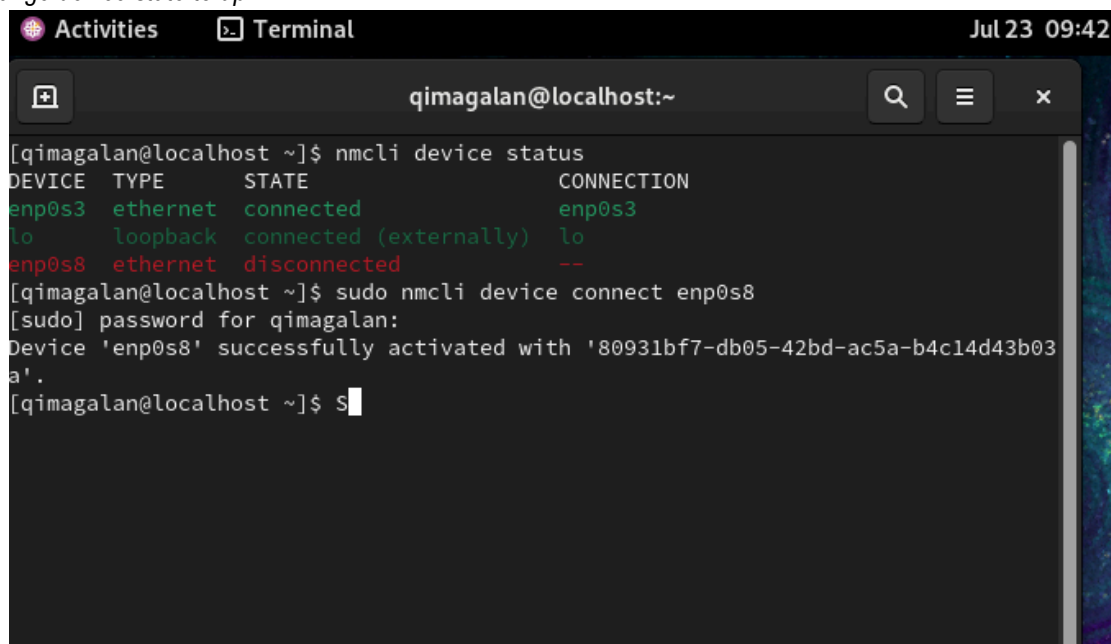
Activities Terminal Jul 23 09:28
qimagalan@localhost:~
[qimagalan@localhost ~]$ sudo systemctl reload sshd
[qimagalan@localhost ~]$

```

Task 3: Copy the Public Key to CentOS

Important Troubleshooting Note: Before proceeding with `ssh-copy-id` command create a new network adapter for CentOS VM with Host-Only Adapter configuration. Another important thing is to verify first with `ifconfig` if the CentOS VM has an assigned IP address. If you do not see any 192.168.x.x IP address then it would not possible to use ssh on the machine as well as `ssh-copy-id` until the fixes below are implemented:

Type **`nmcli device status`** to verify the device status of `enp0s8` (commonly assigned for the second adapter - Host-Only). Notice that the device is disconnected. Simply type the command **`sudo nmcli device connect enp0s8`** to change device state to up.

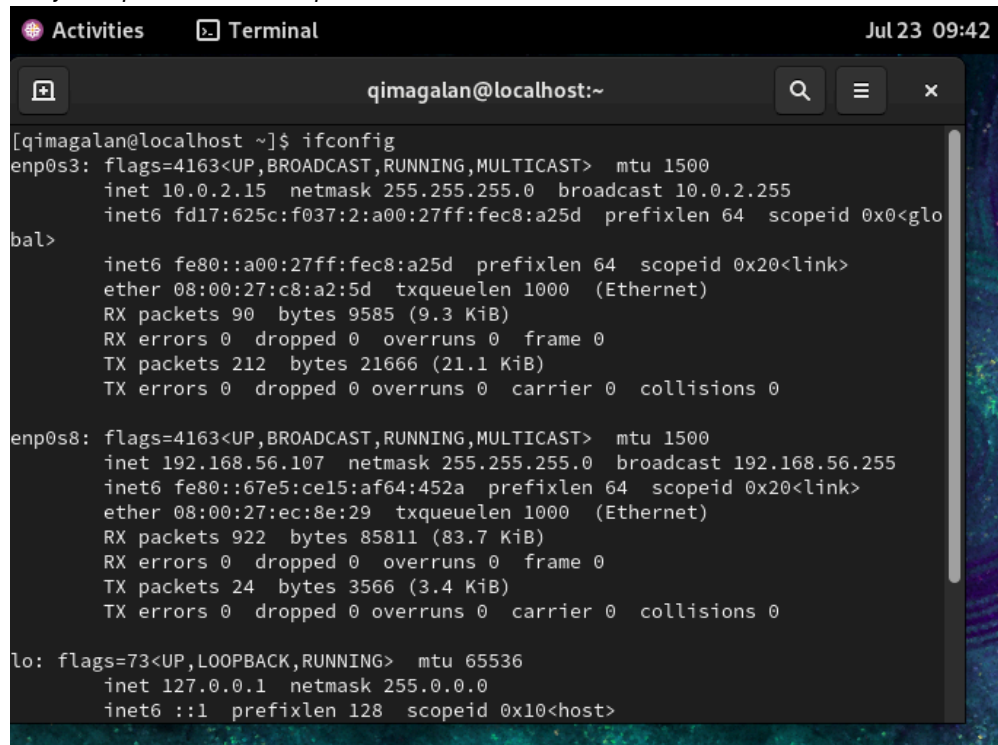


```

Activities Terminal Jul 23 09:42
qimagalan@localhost:~
[qimagalan@localhost ~]$ nmcli device status
DEVICE  TYPE      STATE      CONNECTION
enp0s3  ethernet  connected  enp0s3
lo      loopback  connected (externally)  lo
enp0s8  ethernet  disconnected --
[qimagalan@localhost ~]$ sudo nmcli device connect enp0s8
[sudo] password for qimagalan:
Device 'enp0s8' successfully activated with '80931bf7-db05-42bd-ac5a-b4c14d43b03a'.
[qimagalan@localhost ~]$ s

```

After the fix has been executed, check again with `ifconfig`. `Enp0s8` should now show an ip address 192.168.X.X format. You may now proceed to the steps below.



```

Activities Terminal Jul 23 09:42

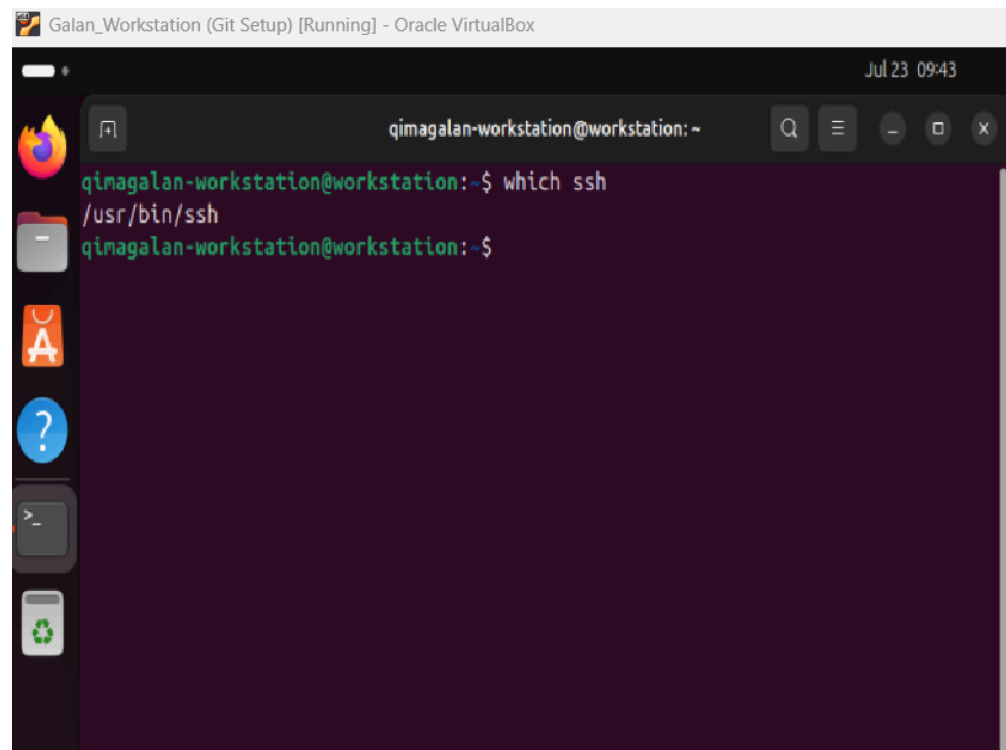
qimagalan@localhost:~

[qimagalan@localhost ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:a00:27ff:fec8:a25d prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:c8:a2:5d txqueuelen 1000 (Ethernet)
    RX packets 90 bytes 9585 (9.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 212 bytes 21666 (21.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.107 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::67e5:ce15:af64:452a prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:ec:8e:29 txqueuelen 1000 (Ethernet)
    RX packets 922 bytes 85811 (83.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 24 bytes 3566 (3.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
  
```

1. Make sure that ssh is installed on the local machine.



```

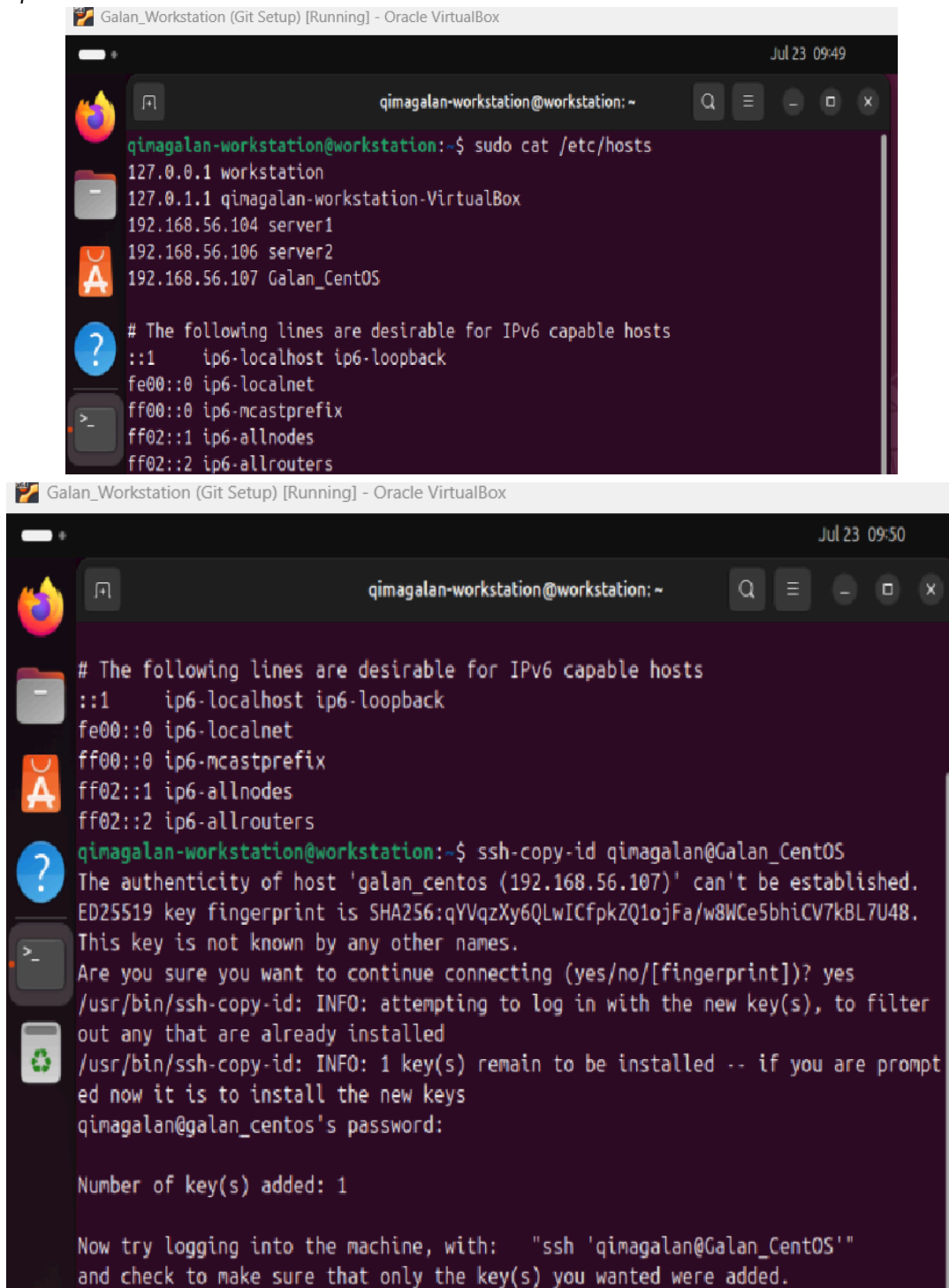
Galan_Workstation (Git Setup) [Running] - Oracle VirtualBox

qimagalan-workstation@workstation:~

qimagalan-workstation@workstation:~$ which ssh
/usr/bin/ssh
qimagalan-workstation@workstation:~$
  
```

2. Using the command `ssh-copy-id`, connect your local machine to CentOS.

Optional: Before using `ssh-copy-id`, it is also convenient to first setup the hostname in the remote CentOS machine, and also add to `/etc/hosts` the ip address corresponding to the remote server name to simplify copying and ssh processes.



The first screenshot shows the contents of the `/etc/hosts` file, which lists several IP addresses and their corresponding hostnames, including the workstation and the CentOS machine.

```
qimagalan-workstation@workstation:~$ sudo cat /etc/hosts
127.0.0.1 workstation
127.0.1.1 qimagalan-workstation-VirtualBox
192.168.56.104 server1
192.168.56.106 server2
192.168.56.107 Galan_CentOS

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

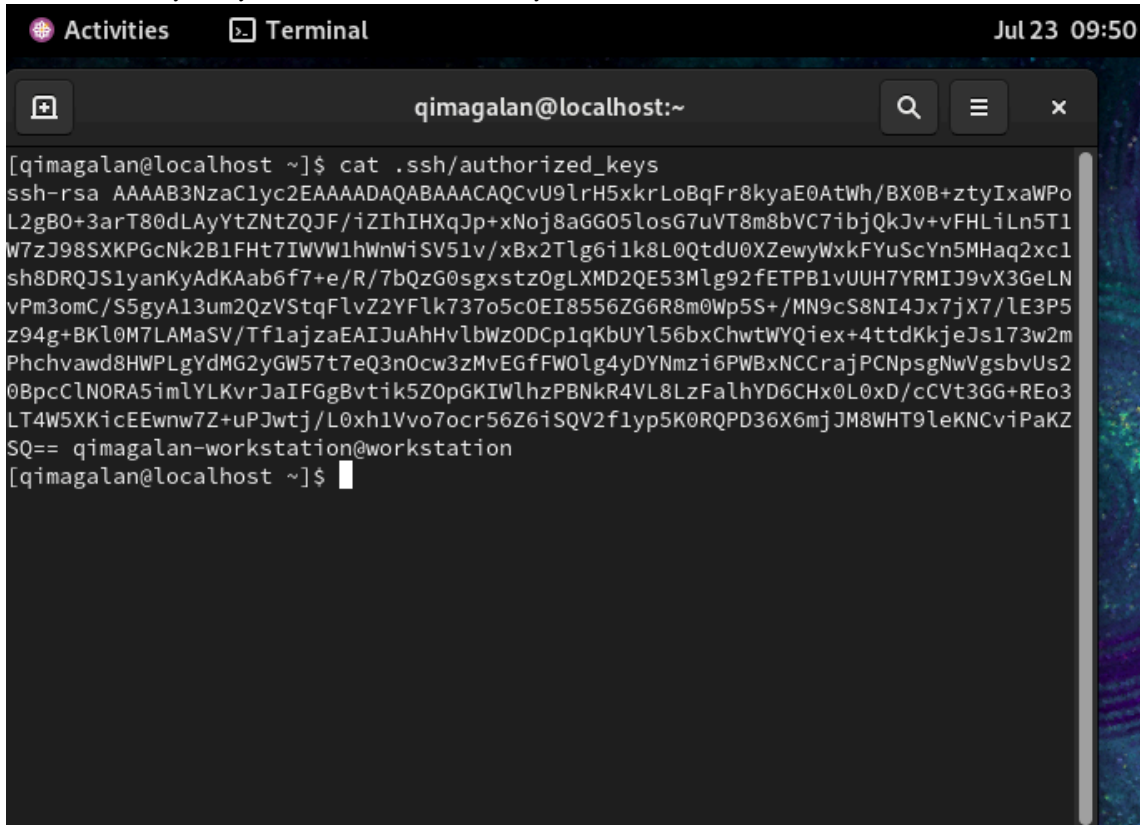
The second screenshot shows the execution of the `ssh-copy-id` command to copy the local SSH key to the CentOS machine. The output indicates that the key was successfully added to the remote machine's `authorized_keys` file.

```
qimagalan-workstation@workstation:~$ ssh-copy-id qimagalan@Galan_CentOS
The authenticity of host 'galan_centos (192.168.56.107)' can't be established.
ED25519 key fingerprint is SHA256:qYVqzXy6QLwICfPkZQ1ojFa/w8WCe5bhiCV7kBL7U48.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
qimagalan@galan_centos's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'qimagalan@Galan_CentOS'"
and check to make sure that only the key(s) you wanted were added.
```

3. On CentOS, verify that you have the `authorized_keys`.

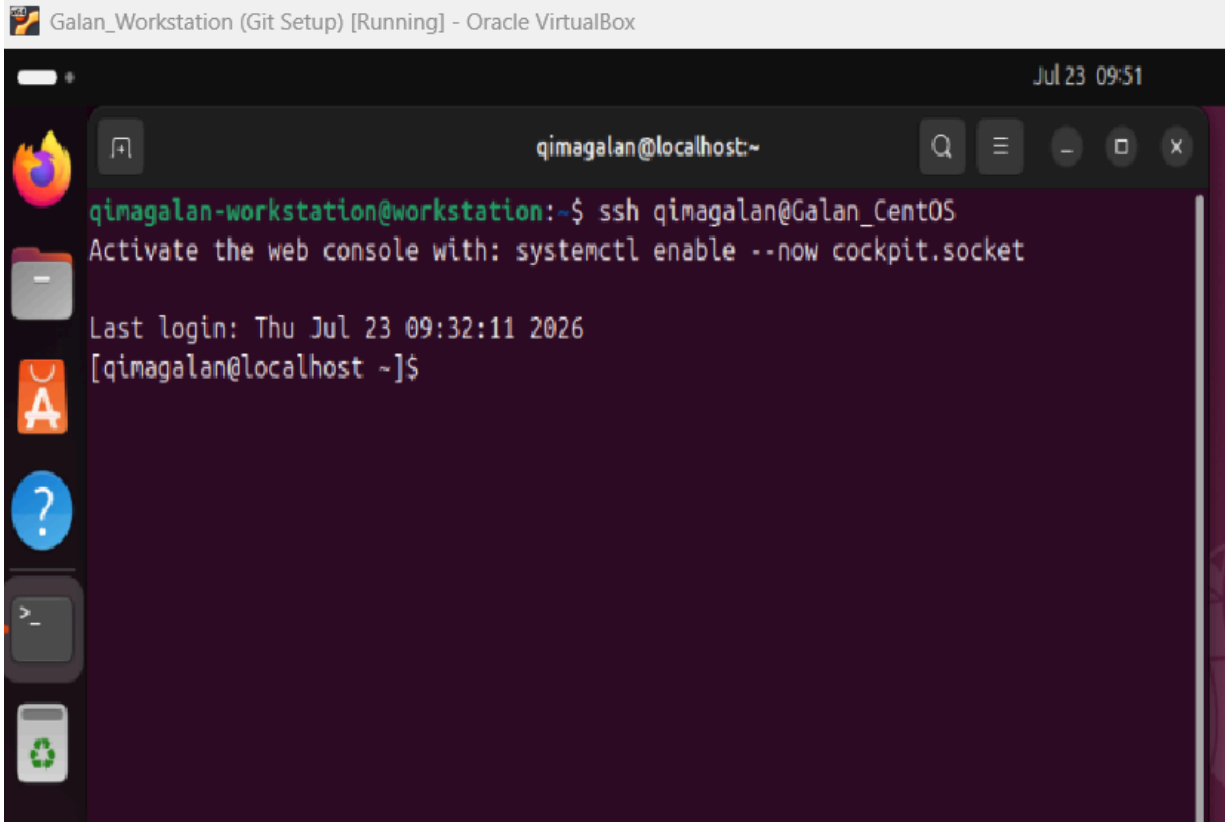


```
qimagalan@localhost:~  
[qimagalan@localhost ~]$ cat .ssh/authorized_keys  
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQACvU9lrH5xkrLoBqFr8kyaE0AtWh/BX0B+ztyIxaWPo  
L2gB0+3arT80dLAyYtZNtZQJF/iZiHlHXqJp+xNoj8aGG05losG7uVT8m8bVC7ibjQkJv+vFHLiLn5T1  
W7zJ98SXKPGcNk2B1Fht7IWVW1hWnWiSV51v/xBx2Tlg6i1k8L0QtdU0XZewyWxkFYuScYn5MHaq2xc1  
sh8DRQJS1yanKyAdKAab6f7+e/R/7bQzG0sgxstz0gLXMD2QE53Mlg92fETPB1vUUH7YRMIJ9vX3GeLN  
vPm3omC/S5gyA13um2QzVstqFlvZ2YFlk737o5c0EI8556ZG6R8m0Wp5S+/MN9cS8NI4Jx7jX7/LE3P5  
z94g+BKL0M7LAMA5V/Tf1ajzaEAIJuAhHvLbWz0DCp1qKbUYl56bxChwtWYQiex+4tttdKkJeJs173w2m  
Phchvawd8HWPLgYdMG2yGW57t7eQ3n0cw3zMvEGfFW0lg4yDYNmzi6PWBxNCCrajPCNpsgNwVgsbvUs2  
0BpcClNORA5imlYLKvrJaIFGgBvtik5Z0pGKIWlhZPBNkR4VL8LzFaLhYD6CHx0L0xD/cCVt3G6+REo3  
LT4W5XKicEEwnw7Z+uPJwtj/L0xh1Vvo7ocr56Z6iSQV2f1yp5K0RQPD36X6mjJM8WHT9leKNCviPaKZ  
SQ== qimagalan-workstation@workstation  
[qimagalan@localhost ~]$
```

Notice how `qimagalan-workstation@workstation` is indicated in the key signifying that it is the ssh key for the specific user we generated in the local machine.

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected.



The screenshot shows a terminal window titled "qimagalan@localhost:~". The prompt is "qimagalan-workstation@workstation:~\$". The user has entered the command "ssh qimagalan@Galan_CentOS". The terminal output shows "Activate the web console with: systemctl enable --now cockpit.socket" and "Last login: Thu Jul 23 09:32:11 2026". The prompt has changed to "[qimagalan@localhost ~]\$".

Notice that no password is asked since I did not set any password and it automatically authenticated through the ssh key. The terminal naming also changed from an Ubuntu Style to a CentOS style.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

Debian based distributions are more friendly for everyday or simply desktop use. If we wanted to practice basic Linux systems administration, debian distros are a great place to start practicing commands as most of them are transferable even for RHEL based distros like Fedora and CentOS. CentOS on the other hand is a gateway not only for basic systems administration but extends towards enterprise-level systems and machines. It is a place where administrators could start learning for actual enterprise distro such as RHEL.

Debian is simple and has an open-source community that regulates and tests software releases and versions which makes it learnable and great for small enterprises. Red Hat however wins in terms of systems that would need critical performance which is why large enterprises often operate in RHEL, and it also uses CentOS as a testing field for patches to make sure that it is stable when released into RHEL. Red Hat is used in systems requiring strict compliance, security, and data confidentiality (Meghwani M., 2025).

2. What are the main differences between Debian and Red Hat Linux distributions?

Debian is primarily open-source in its software and therefore developers and programmers of system administration could download and make modifications in tools that would suit their needs or needs of the community (Meghwani M., 2025 - 'Software Freedom'). Debian capitalizes on its community support. On the other hand, enterprises largely lean on Red Hat distros as it is a half community-driven and half of it also has commercial support which allows companies to not worry about long term maintenance and support as RHEL makes adequate income to sustain frequent updates for security and maintenance as well as the formal documentation of features. It also offers support for the latest leading hardware and software vendors (Meghwani M., 2025 - 'Software Freedom'). Simply put Debian relies on the community for debugging, but for Red Hat you have a commercial company backing the OS debugging and security updates that is critical for company servers.

Reference:

Meghwani, M. (2025, November 7). *Debian vs RHEL: Key differences in Linux distributions*. ServerAvatar. <https://serveravatar.com/debian-vs-rhel/>

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Claude

Purpose of Use:

For the IP address troubleshooting of the enp0s8 host-only adapter in Task 3

Extent of Use:

The AI was only used for finding the specific command for troubleshooting. As good practice I always check the ifconfig to verify whether I could ssh properly into another device, and upon noticing the missing IP address I checked with AI the possible misconfigurations.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.